Nihar Gaddam

nihar.ignis@gmail.com ● 412-916-4606 ● https://www.linkedin.com/in/nihar-gaddam/

EDUCATION

Carnegie Mellon University Pittsburgh, PA
Master of Science in Computer Engineering – Applied Advanced Study Dec 2023

- **GPA:** 3.75/4.00
- **Relevant Coursework:** Distributed Systems, Cloud Computing, Foundations of Computer Systems, Computer Vision, Machine learning for Signal Processing, Introduction to Deep learning

Sreenidhi Institute of Science and Technology Hyderabad, India
Bachelor of Technology in Electronics and Communication Engineering June 2020

- **GPA:** 9.39/10.00
- Minor in Computer Science and Engineering

SKILLS

Languages: Go, Rust, Python, C++, Java
IAAS: AWS, Azure, GCP
Container Platform: Docker, Kubernetes
CI/CD: GitHub Actions, Apache Airflow
IAC & Config Management: Terraform, Puppet
Storage: HBase, HDFS, Zookeeper, MySQL, MongoDB, Neo4j, Postgres
Frameworks/Tools: PyTorch, Spark, Kafka, Vertex AI
Monitoring and Observability: Prometheus, Grafana, Splunk, OpenTelemetry, SignalFx

EXPERIENCE

Cisco Systems Inc. Pittsburgh, PA (Aug 2024 - Present)
Software Development Engineer

- **Backend Engineering & API Development**
  - Designed and maintained features for CO2 (Cloud Orchestration 2), a Go-based Kubernetes operator managing the full lifecycle of tens of thousands of Splunk Cloud stacks, and ACS (Admin Config Service), a customer-facing configuration API.

- **System Monitoring & Performance Optimization**
  - Continuously monitored and optimized Kubernetes cluster deployments, leveraging Skynet logs to identify and resolve performance bottlenecks; reduced technical debt through regular code cleanup and targeted refactors across development, staging, and production environments.

- **Key Projects and Achievements**
  - **Cloud Preview Builds**: Replaced a multi-month per-program manual provisioning process with a declarative, self-serve preview stack path adopted by other orgs like AI Canvas and Federated Analytics 2.0 — ~100 active preview stacks per month in production with zero per-preview engineering coordination. Authored the accepted RFC and led the CO2 workstream end-to-end across Cloudworks, CICD, VOC, TechOps, and Puppet/ACM teams.
  - **SLO Instrumentation & Observability**: Built CO2's first time-series view of provisioning success and latency across a Splunk Cloud environments via custom Prometheus metrics and an OpenTelemetry collector sidecar forwarding to SignalFx. Shipped multiple SLO-based detectors that proactively caught cross-team incidents that previously surfaced only via customer raised incidents.
  - **XRDR Termination Safety**: Identified and fixed a latent ordering bug where TTL-based teardown could destroy an XRDR Primary stack while its dependent Secondary was still live. Authored the accepted RFC and rolled out modified Reconciler design to block Primary teardown until all Secondaries are confirmed terminated.
  - **Automated Trial Stack cleanup**: Identified that trial stacks were accumulating in USED status indefinitely due to missed deletion signals from Salesforce. Implemented and rolled out an automated deletion operator across all environments that terminates trial stacks that have remained in USED status beyond a set deadline. The rollout reclaimed 2,385 stale stacks across environments, directly reducing unnecessary cloud spend fleet-wide.
  - **P2 Integration & Encryption-at-Rest**: Integrated CO2 with the product-packaging service (P2) via IAC authentication and ephemeral tokens, eliminating 140+ manual ticket requests annually and ensuring all new stacks are provisioned with the latest Splunk release. Led EAR enablement by default for all new stacks and audited 3,000+ existing stacks, achieving 100% adoption.
  - **XRDR Failover Read Access**: Enabled customer read access on secondary XRDR stacks during failovers, reducing incident resolution time by 30% and improving business continuity for 800+ customers.
  - **Security & Reliability**: Identified allowlists lacking security-group filtering and implemented blocking measures, reducing related incident tickets by 24%. Propagated certificate rotation status to customer-visible stack CRs, cutting certificate-related support queries by 23% for 8,000+ users.
  - **Azure Reliability / CO2<>Windex**: Authored an RFC and implemented the rollout to offload Azure subscription pool management from ad-hoc Terraform logic to the Windex service at the CO2 Network reconciler layer, making acquisition failures observable and retryable as Kubernetes conditions.
  - **Vault-Template Token Automation**: Automated legacy authentication token refresh across multiple services using vault-template, increasing token management efficiency by ~40% and eliminating manual intervention for 15 service accounts.

Accenture Solutions Pvt. Ltd.
Application Development Associate Hyderabad, India (Dec 2020 – Oct 2021)

- **Cloud Infrastructure Management**
  - Migrated client infrastructure from on-premises to AWS and Azure, improving scalability and reducing operational costs by 20%; maintained development and production environments, reducing downtime by up to 40%.
  - Executed high-risk production changes with minimal disruption, maintaining a 99.8% success rate in change management.
- **Data Analysis and Reporting**
  - Leveraged Power BI for real-time server performance trend analysis, reducing response time to performance issues by 30%; managed ETL dataflows from Azure SQL, SharePoint, Redshift, MongoDB, and Snowflake into BI reporting pipelines

PROJECTS

Personal

SVID-Exchange | *Go* 🔗 Mar 2026

- Built a Zero Trust token exchange service that converts SPIFFE SVIDs into scoped, short-lived ES256 JWTs over mTLS, eliminating static shared secrets for service-to-service authentication.

Saga-Conductor | *Go* 🔗 Mar 2026

- Built a lightweight saga-pattern orchestrator for distributed transactions: callers define each step with a forward action and a compensating action, and the conductor executes them in order, automatically rolling back completed steps in reverse on failure.

Academic

Distributed Backend Storage for Multiplayer Environments | *Go* Dec 2023

- Created a distributed key-value storage system utilizing replication, partitioning, and concurrency mechanisms to support a globally distributed multiplayer online game, ensuring data consistency and low-latency access across multiple backend servers.

Analysis of Twitter user graph with Spark | *Scala* Nov 2023

- Proficiently applied Apache Spark programs on Azure HDInsight to analyze Twitter social graph; implemented PageRank algorithm to identify influential users and optimized the runtime from 60 to 28 minutes, enhancing performance.

Advanced resource scaling using Cloud SDKs | *Python* Sep 2023

- Implemented horizontal scaling with Boto3 AWS SDK, reducing response times from 30 to 8 minutes; increased RPS from 8K request workloads to 40; configured Elastic Load Balancers, Auto Scaling Groups, and Terraform for efficient deployment.

CERTIFICATIONS

- Certified Kubernetes Application Developer (CKAD), Cloud Native Computing Foundation (CNCF) – 2024 🔗